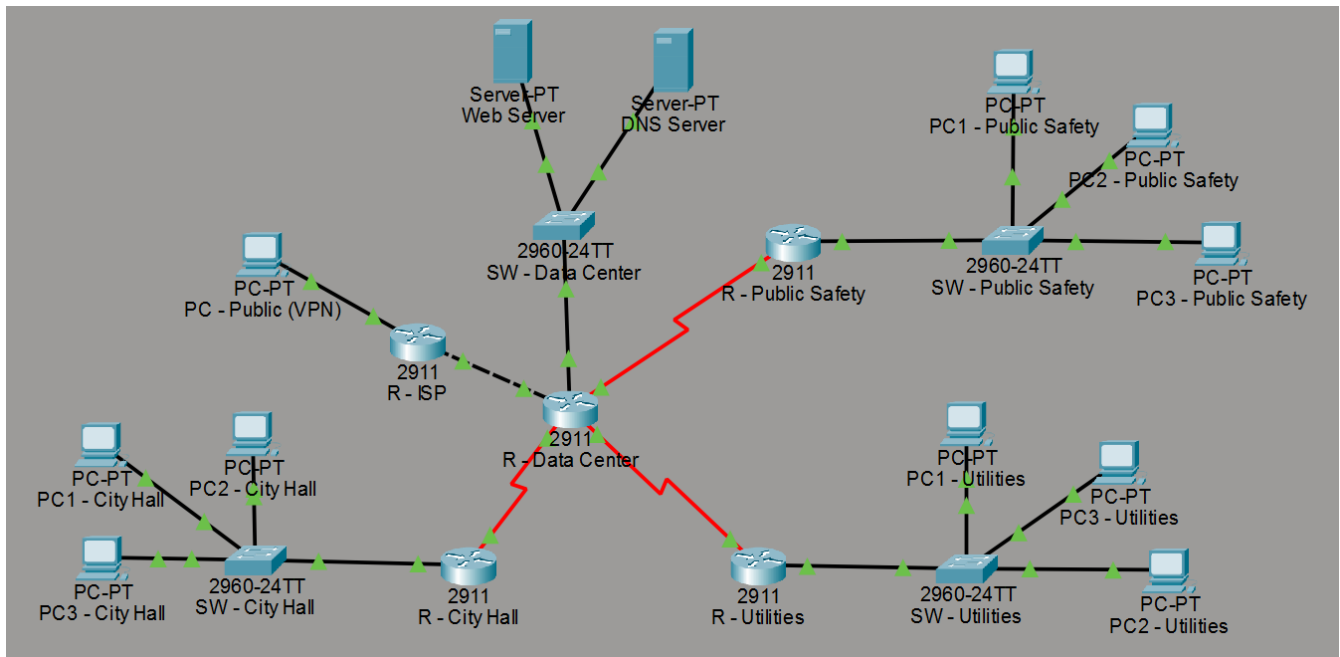


The Smart City Integrated Infrastructure

1. Network Design

A. Overall Network



B. Hardware, Software, and Cabling Specifications

Component	Device/Model	Specifications / IOS Version
Routers	Cisco 2911	Cisco IOS Software, Version 15.1(4)M4 (SecurityK9 Bundle)
Switches	Cisco 2960-24TT	Cisco IOS Software, Version 15.0(2)
Servers	Generic PT-Server	HTTP/DNS Services Enabled
Cabling (LAN)	Copper Straight-Through	PC to Switch (Access), Switch to Router (Trunk)
Cabling (WAN)	Serial DCE/DTE	High-speed Serial links for Inter-Router connectivity
Cabling (ISP)	Gigabit Ethernet	Router to ISP Cloud (Copper Straight-Through)

C. Protocol Documentation Map

Service / Protocol	Physical Location	Logical Interface / Scope
DHCP Server	R-DataCenter	Centralized Pool for 192.168.10.0/26 (Utilities)
DHCP Relay	R-Utilities	ip helper-address configured on Sub-interface G0/0.10
NAT (Static)	R-DataCenter	Private 192.168.10.99 mapped to Public 200.1.1.10
NAT (PAT)	R-DataCenter	Overload on GigabitEthernet0/1 for all internal subnets
VPN Tunnel Start	R-DataCenter	Interface GigabitEthernet0/1 (Local: 200.1.1.2)
VPN Tunnel End	R-ISP	Interface GigabitEthernet0/0 (Remote: 200.1.1.1)
Routing Protocol	All Routers	Static Routing

2. VLSM Calculation

Our specific Major Network (IP block) is the standard private Class C network **192.168.10.0/24**.

For the **Data Center**, a specific host count wasn't given, but it needs to accommodate an internal DNS server, a Public Web Server, and a VPN concentrator. I have estimated **10 hosts** for this sector, which fits efficiently into a /28 block.

- Sort by Size:** You must always subnet starting with the largest requirement.
 - Utilities (60)** needs a block size of 64 (2^6), which is a /26 mask.
 - City Hall (30)** needs a block size of 32 (2^5), which is a /27 mask.
 - Public Safety (14)** needs a block size of 16 (2^4), which is a /28 mask.
- Place the Blocks:**
 - Utilities** takes the first spot: .0 to .63.
 - City Hall** starts immediately after: .64 to .95.
 - Data Center** (Servers) fits in the next slot. Since we estimated 10 hosts, a /28 (16 hosts) is perfect. It occupies .96 to .111.
 - Public Safety** fits exactly in the next /28 block: .112 to .127.
- WAN Links:**
 - Routers need to talk to each other. Point-to-Point links only need 2 usable IPs. A /30 mask (block size of 4) is standard for this.
 - These are placed at the end of the range (.128, .132, .136) to keep the user subnets grouped at the beginning.

Subnet Name	Network ID	Subnet Mask	First IP	Last IP	Broadcast IP
Utilities (Largest)	192.168.10.0	255.255.255.192	192.168.10.1	192.168.10.62	192.168.10.63
City Hall (2nd)	192.168.10.64	255.255.255.224	192.168.10.65	192.168.10.94	192.168.10.95
Data Center	192.168.10.96	255.255.255.240	192.168.10.97	192.168.10.110	192.168.10.111
Public Safety	192.168.10.112	255.255.255.240	192.168.10.113	192.168.10.126	192.168.10.127
WAN Link (Center ↔ Utilities)	192.168.10.128	255.255.255.252	192.168.10.129	192.168.10.130	192.168.10.131
WAN Link (Center ↔ City Hall)	192.168.10.132	255.255.255.252	192.168.10.133	192.168.10.134	192.168.10.135
WAN Link (Center ↔ Public Safety)	192.168.10.136	255.255.255.252	192.168.10.137	192.168.10.138	192.168.10.139

3. Addressing Table

A. Data Center Router

R-DataCenter#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	192.168.10.97	YES	manual	up	up
GigabitEthernet0/0.100	unassigned	YES	unset	up	up
GigabitEthernet0/1	200.1.1.2	YES	NVRAM	up	up
GigabitEthernet0/2	unassigned	YES	NVRAM	administratively down	down
Serial0/0/0	192.168.10.129	YES	manual	up	up
Serial0/0/1	192.168.10.133	YES	manual	up	up
Serial0/1/0	192.168.10.137	YES	manual	up	up
Serial0/1/1	unassigned	YES	unset	down	down
Virtual-Template1	200.1.1.2	YES	unset	down	down
Virtual-Access1	unassigned	YES	unset	down	down
Vlan1	unassigned	YES	unset	administratively down	down

B. City Hall Router

R-CityHall#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	manual	up	up
GigabitEthernet0/0.20	192.168.10.65	YES	manual	up	up
GigabitEthernet0/1	unassigned	YES	NVRAM	administratively down	down
GigabitEthernet0/2	unassigned	YES	NVRAM	administratively down	down
Serial0/0/0	192.168.10.134	YES	manual	up	up
Serial0/0/1	unassigned	YES	unset	down	down
Vlan1	unassigned	YES	unset	administratively down	down

C. Utilities Router

R-Utilities#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	manual	up	up
GigabitEthernet0/0.10	192.168.10.1	YES	manual	up	up
GigabitEthernet0/1	unassigned	YES	NVRAM	administratively down	down
GigabitEthernet0/2	unassigned	YES	NVRAM	administratively down	down
Serial0/0/0	192.168.10.130	YES	manual	up	up
Serial0/0/1	unassigned	YES	unset	down	down
Vlan1	unassigned	YES	unset	administratively down	down

D. Public Safety Router

R-PublicSafety#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	manual	up	up
GigabitEthernet0/0.30	192.168.10.113	YES	manual	up	up
GigabitEthernet0/1	unassigned	YES	NVRAM	administratively down	down
GigabitEthernet0/2	unassigned	YES	NVRAM	administratively down	down
Serial0/0/0	192.168.10.138	YES	manual	up	up
Serial0/0/1	unassigned	YES	unset	down	down
Vlan1	unassigned	YES	unset	administratively down	down

E. ISP Router

R-ISP#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	200.1.1.1	YES	NVRAM	up	up
GigabitEthernet0/1	200.1.20.1	YES	NVRAM	up	up
GigabitEthernet0/2	unassigned	YES	NVRAM	administratively down	down
Loopback0	8.8.8.8	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

4. DHCP

R-DataCenter#show ip dhcp pool

Pool CITY_HALL_POOL :

Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 30
Leased addresses : 3
Excluded addresses : 7
Pending event : none

1 subnet is currently in the pool

Current index	IP address range	Leased/Excluded/Total
192.168.10.65	192.168.10.65 - 192.168.10.94	3 / 7 / 30

Pool PUBLIC_SAFETY_POOL :

Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 14
Leased addresses : 3
Excluded addresses : 7
Pending event : none

1 subnet is currently in the pool

Current index	IP address range	Leased/Excluded/Total
192.168.10.113	192.168.10.113 - 192.168.10.126	3 / 7 / 14

Pool UTILITIES_POOL :

Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 62
Leased addresses : 3
Excluded addresses : 7
Pending event : none

1 subnet is currently in the pool

Current index	IP address range	Leased/Excluded/Total
192.168.10.1	192.168.10.1 - 192.168.10.62	3 / 7 / 62

```
C:\>ipconfig /all
```

```
FastEthernet0 Connection:(default port)
```

```
Connection-specific DNS Suffix...:
```

```
Physical Address.....: 000A.F3AC.8D44
```

```
Link-local IPv6 Address.....: FE80::20A:F3FF:FEAC:8D44
```

```
IPv6 Address.....: ::
```

```
IPv4 Address.....: 192.168.10.78
```

```
Subnet Mask.....: 255.255.255.224
```

```
Default Gateway.....: ::
```

```
192.168.10.65
```

```
DHCP Servers.....: 192.168.10.133
```

```
DHCPv6 IAID.....:
```

```
DHCPv6 Client DUID.....: 00-01-00-01-61-AB-4D-A4-00-0A-F3-AC-8D-44
```

```
DNS Servers.....: ::
```

```
192.168.10.99
```

```
Bluetooth Connection:
```

```
Connection-specific DNS Suffix...:
```

```
Physical Address.....: 00E0.A399.2A0E
```

```
Link-local IPv6 Address.....: ::
```

```
IPv6 Address.....: ::
```

```
IPv4 Address.....: 0.0.0.0
```

```
Subnet Mask.....: 0.0.0.0
```

```
Default Gateway.....: ::
```

```
0.0.0.0
```

```
DHCP Servers.....: 0.0.0.0
```

```
DHCPv6 IAID.....:
```

```
DHCPv6 Client DUID.....: 00-01-00-01-61-AB-4D-A4-00-0A-F3-AC-8D-44
```

```
DNS Servers.....: ::
```

```
192.168.10.99
```

5. DNS Server

```
C:\>ping www.smartcity.gov
```

```
Pinging 192.168.10.98 with 32 bytes of data:
```

```
Reply from 192.168.10.98: bytes=32 time=12ms TTL=126
```

```
Reply from 192.168.10.98: bytes=32 time=2ms TTL=126
```

```
Reply from 192.168.10.98: bytes=32 time=11ms TTL=126
```

```
Reply from 192.168.10.98: bytes=32 time=2ms TTL=126
```

```
Ping statistics for 192.168.10.98:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
    Approximate round trip times in milli-seconds:
```

```
        Minimum = 2ms, Maximum = 12ms, Average = 6ms
```

6. VLAN

A. Utilities Switch

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/2
10	Utilities	active	Fa0/1, Fa0/2, Fa0/3
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

B. Public Safety Switch

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/2
30	Public_Safety	active	Fa0/1, Fa0/2, Fa0/3
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

C. City Hall Switch

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/2
20	City_Hall	active	Fa0/1, Fa0/2, Fa0/3
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

7. NAT (Static/PAT)

```
R-DataCenter#show ip nat translations
```

Pro	Inside global	Inside local	Outside local	Outside global
icmp	200.1.1.2:1024	192.168.10.13:1	8.8.8.8:1	8.8.8.8:1024
icmp	200.1.1.2:1025	192.168.10.13:2	8.8.8.8:2	8.8.8.8:1025
icmp	200.1.1.2:1026	192.168.10.13:3	8.8.8.8:3	8.8.8.8:1026
icmp	200.1.1.2:1027	192.168.10.13:4	8.8.8.8:4	8.8.8.8:1027
icmp	200.1.1.2:1	192.168.10.122:1	8.8.8.8:1	8.8.8.8:1
icmp	200.1.1.2:2	192.168.10.122:2	8.8.8.8:2	8.8.8.8:2
icmp	200.1.1.2:3	192.168.10.122:3	8.8.8.8:3	8.8.8.8:3
icmp	200.1.1.2:4	192.168.10.122:4	8.8.8.8:4	8.8.8.8:4
---	200.1.1.10	192.168.10.99	---	---

```
R-DataCenter#show ip nat statistics
```

```
Total translations: 9 (1 static, 8 dynamic, 8 extended)
```

```
Outside Interfaces: GigabitEthernet0/1
```

```
Inside Interfaces: Serial0/0/0 , Serial0/0/1 , Serial0/1/0 , GigabitEthernet0/0.100
```

```
Hits: 15 Misses: 16
```

```
Expired translations: 8
```

```
Dynamic mappings:
```

8. VPN

A. Data Center Router

```
R-DataCenter#show crypto isakmp sa
```

```
IPv4 Crypto ISAKMP SA
```

dst	src	state	conn-id	slot	status
200.1.1.1	200.1.1.2	QM_IDLE	1085	0	ACTIVE

```
IPv6 Crypto ISAKMP SA
```

```
R-DataCenter#show crypto ipsec sa
```

```
interface: GigabitEthernet0/1
```

```
  Crypto map tag: MYMAP, local addr 200.1.1.2
```

```
protected vrf: (none)
```

```
local ident (addr/mask/prot/port): (192.168.10.0/255.255.255.0/0/0)
```

```
remote ident (addr/mask/prot/port): (200.1.20.0/255.255.255.0/0/0)
```

```
current_peer 200.1.1.1 port 500
```

```
  PERMIT, flags={origin_is_acl,}
```

```
#pkts encaps: 3, #pkts encrypt: 3, #pkts digest: 0
```

```
#pkts decaps: 2, #pkts decrypt: 2, #pkts verify: 0
```

```
#pkts compressed: 0, #pkts decompressed: 0
```

```
#pkts not compressed: 0, #pkts compr. failed: 0
```

```
#pkts not decompressed: 0, #pkts decompress failed: 0
```

```
#send errors 1, #recv errors 0
```

```
local crypto endpt.: 200.1.1.2, remote crypto endpt.:200.1.1.1
```

```
path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/1
```

```
current outbound spi: 0x199E0E9D(429788829)
```



```

inbound ah sas:

inbound pcp sas:

outbound esp sas:
  spi: 0x3764DE83(929357443)
    transform: esp-aes esp-sha-hmac ,
    in use settings ={Tunnel, }
    conn id: 2010, flow_id: FPGA:1, crypto map: ISP_MAP
    sa timing: remaining key lifetime (k/sec): (4525504/3503)
    IV size: 16 bytes
    replay detection support: N
    Status: ACTIVE

outbound ah sas:

outbound pcp sas:

```

B. ISP Router

```

R-ISP#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status
200.1.1.2    200.1.1.1    QM_IDLE        1096      0  ACTIVE

IPv6 Crypto ISAKMP SA

R-ISP#show crypto ipsec sa

interface: GigabitEthernet0/0
  Crypto map tag: ISP_MAP, local addr 200.1.1.1

protected vrf: (none)
local  ident (addr/mask/prot/port): (200.1.20.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.10.0/255.255.255.0/0/0)
current_peer 200.1.1.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 2, #pkts encrypt: 2, #pkts digest: 0
#pkts decaps: 3, #pkts decrypt: 3, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0

local crypto endpt.: 200.1.1.1, remote crypto endpt.:200.1.1.2
path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/0
current outbound spi: 0x3764DE83(929357443)

inbound esp sas:
  spi: 0x199E0E9D(429788829)
    transform: esp-aes esp-sha-hmac ,
    in use settings ={Tunnel, }
    conn id: 2009, flow_id: FPGA:1, crypto map: ISP_MAP
    sa timing: remaining key lifetime (k/sec): (4525504/3503)
    IV size: 16 bytes
    replay detection support: N
    Status: ACTIVE

```


inbound esp sas:
spi: 0x3764DE83(929357443)
transform: esp-aes esp-sha-hmac ,
in use settings ={Tunnel, }
conn id: 2009, flow_id: FPGA:1, crypto map: MYMAP
sa timing: remaining key lifetime (k/sec): (4525504/3576)
IV size: 16 bytes
replay detection support: N
Status: ACTIVE

inbound ah sas:

inbound pcg sas:

outbound esp sas:
spi: 0x199E0E9D(429788829)
transform: esp-aes esp-sha-hmac ,
in use settings ={Tunnel, }
conn id: 2010, flow_id: FPGA:1, crypto map: MYMAP
sa timing: remaining key lifetime (k/sec): (4525504/3576)
IV size: 16 bytes
replay detection support: N
Status: ACTIVE

outbound ah sas:

outbound pcg sas:

9. Static Routing

A. Data Center Router

```
R-DataCenter#show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP  
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP  
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area  
* - candidate default, U - per-user static route, o - ODR  
P - periodic downloaded static route
```

```
Gateway of last resort is 200.1.1.1 to network 0.0.0.0
```

```
192.168.10.0/24 is variably subnetted, 11 subnets, 5 masks  
S    192.168.10.0/26 is directly connected, Serial0/0/0  
      [1/0] via 192.168.10.130  
S    192.168.10.64/27 is directly connected, Serial0/0/1  
      [1/0] via 192.168.10.134  
C    192.168.10.96/28 is directly connected, GigabitEthernet0/0  
L    192.168.10.97/32 is directly connected, GigabitEthernet0/0  
S    192.168.10.112/28 is directly connected, Serial0/1/0  
      [1/0] via 192.168.10.138  
C    192.168.10.128/30 is directly connected, Serial0/0/0  
L    192.168.10.129/32 is directly connected, Serial0/0/0  
C    192.168.10.132/30 is directly connected, Serial0/0/1  
L    192.168.10.133/32 is directly connected, Serial0/0/1  
C    192.168.10.136/30 is directly connected, Serial0/1/0  
L    192.168.10.137/32 is directly connected, Serial0/1/0  
200.1.1.0/24 is variably subnetted, 2 subnets, 2 masks  
C    200.1.1.0/30 is directly connected, GigabitEthernet0/1  
L    200.1.1.2/32 is directly connected, GigabitEthernet0/1  
S*   0.0.0.0/0 [1/0] via 200.1.1.1
```

B. City Hall Router

```
R-CityHall#show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP  
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP  
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area  
* - candidate default, U - per-user static route, o - ODR  
P - periodic downloaded static route
```

```
Gateway of last resort is 192.168.10.133 to network 0.0.0.0
```

```
192.168.10.0/24 is variably subnetted, 4 subnets, 3 masks  
C    192.168.10.64/27 is directly connected, GigabitEthernet0/0.20  
L    192.168.10.65/32 is directly connected, GigabitEthernet0/0.20  
C    192.168.10.132/30 is directly connected, Serial0/0/0  
L    192.168.10.134/32 is directly connected, Serial0/0/0  
S*   0.0.0.0/0 [1/0] via 192.168.10.133
```

C. Utilities Router

```
R-Utilities#show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP  
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP  
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area  
* - candidate default, U - per-user static route, o - ODR  
P - periodic downloaded static route
```

```
Gateway of last resort is 192.168.10.129 to network 0.0.0.0
```

```
192.168.10.0/24 is variably subnetted, 4 subnets, 3 masks  
C    192.168.10.0/26 is directly connected, GigabitEthernet0/0.10  
L    192.168.10.1/32 is directly connected, GigabitEthernet0/0.10  
C    192.168.10.128/30 is directly connected, Serial0/0/0  
L    192.168.10.130/32 is directly connected, Serial0/0/0  
S*   0.0.0.0/0 [1/0] via 192.168.10.129
```

D. Public Safety Router

```
R-PublicSafety#show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP  
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP  
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area  
* - candidate default, U - per-user static route, o - ODR  
P - periodic downloaded static route
```

```
Gateway of last resort is 192.168.10.137 to network 0.0.0.0
```

```
192.168.10.0/24 is variably subnetted, 4 subnets, 3 masks  
C    192.168.10.112/28 is directly connected, GigabitEthernet0/0.30  
L    192.168.10.113/32 is directly connected, GigabitEthernet0/0.30  
C    192.168.10.136/30 is directly connected, Serial0/0/0  
L    192.168.10.138/32 is directly connected, Serial0/0/0  
S*   0.0.0.0/0 [1/0] via 192.168.10.137
```

E. ISP Router

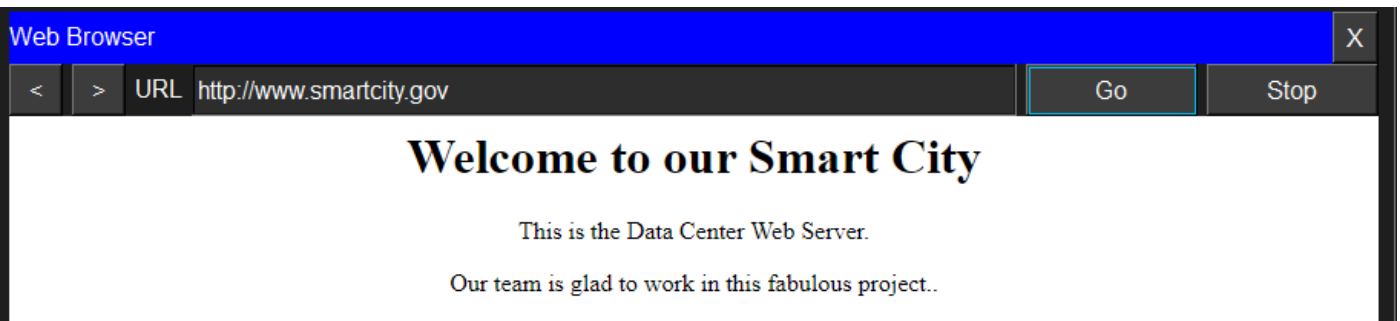
```
R-ISP#show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
      8.0.0.0/32 is subnetted, 1 subnets
C       8.8.8.8/32 is directly connected, Loopback0
S      192.168.10.0/24 [1/0] via 200.1.1.2
      200.1.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       200.1.1.0/30 is directly connected, GigabitEthernet0/0
L       200.1.1.1/32 is directly connected, GigabitEthernet0/0
      200.1.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       200.1.20.0/24 is directly connected, GigabitEthernet0/1
L       200.1.20.1/32 is directly connected, GigabitEthernet0/1
```

10. Web Server



11. ICMP Testing

PDU List Window								
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
	Successful	PC2 - City Hall	PC2 - Utilities	ICMP		0.000	N	0
	Successful	PC3 - Public Safety	PC3 - City Hall	ICMP		0.000	N	1
	Successful	PC2 - Utilities	PC3 - Public Safety	ICMP		0.000	N	2

```
C:\>ping 192.168.10.11

Pinging 192.168.10.11 with 32 bytes of data:

Reply from 192.168.10.11: bytes=32 time=10ms TTL=125
Reply from 192.168.10.11: bytes=32 time=10ms TTL=125
Reply from 192.168.10.11: bytes=32 time=15ms TTL=125
Reply from 192.168.10.11: bytes=32 time=13ms TTL=125

Ping statistics for 192.168.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 15ms, Average = 12ms
```